

OISHY SAHA

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SUMMARY

Ph.D. student in Electrical and Computer Engineering with hands-on experience taking ML models from research to real-time deployment including multimodal anomaly detection, speech/audio processing, model compression, and FPGA/edge inference. Author of 7+ peer-reviewed publications. Seeking a Fall 2026/Summer 2027 internship in machine learning, embedded AI, or signal processing.

EDUCATION

University of Maryland, College Park

Aug 2023 – Expected (May 2028)

Ph.D. in Electrical and Computer Engineering

Graduate Research Assistant, DSPCAD Lab , Advisor: Dr. Shuvra Bhattacharyya

Bangladesh University of Engineering and Technology (BUET)

Feb 2017 – May 2022

B.Sc. in Electrical and Electronic Engineering

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Bash, MATLAB, Verilog HDL, Assembly language

AI/ML Frameworks: PyTorch, TensorFlow, OpenCV, Weights and Biases, Hugging Face

Model Deployment & Optimization: TFLite, ONNX, Pruning, Quantization

Embedded/Hardware Tools: Xilinx Vivado, Altera Quartus, FPGA programming, Arduino, Raspberry Pi

Tools & Platforms: Linux, Git, GitHub, GitLab, VS Code, ns-3, Simulink, Proteus, PSpice

RESEARCH EXPERIENCE

1. Real-Time Image Analysis for Additive Manufacturing (RIAM):

- Improved defect-detection recall by 10% and AUROC/AUPR by 5% by designing semi-supervised pipelines leveraging normal melt pool data.
- Built multimodal anomaly-detection pipelines for laser powder bed fusion (LPBF) fusing melt-pool images, scan-strategy data, and layer images under limited labeled defect annotations.
- Developed knowledge-distilled teacher-student models for real-time melt-pool analysis with reduced model complexity.
- Deploying lightweight inference on Kintex-UltraScale FPGA hardware with an inference target of 2 ms per image.

2. Real-Time Wireless Communication for Industrial Networks (RWCIN):

- Generated labeled cyber-attack datasets spanning normal, single-attack, sequential multi-attack, and simultaneous mixed-attack scenarios comprising 1M samples across 6 attack types.
- Developed ML pipelines for anomaly detection, attack classification, multi-label detection, sequential-attack tracking, and unseen-attack generalization on industrial wireless traffic.
- Deploying cyberattack detection models on edge devices.

3. Maryland-Nebraska Video Analytics for Public Safety (MNVP):

- Curated and annotated 20+ hours audio-visual body-worn camera footage from the Baltimore Police Department (BPD) to build a labeled dataset for egocentric activity recognition.
- Built speech emotion recognition (SER) pipelines for continuous valence/arousal regression using pretrained audio models including VGGish, wav2vec 2.0, emotion2vec, and WavLM.
- Simulated cross-domain experiments across RAVDESS, MSP-Podcast, and noisy real-world BPD audio for robust emotion recognition under domain shift, improving CCC by 40% over baseline.

4. Undergraduate Thesis: Automatic Emotion Recognition Using EEG Signal

- Developed deep learning models (CNN, BiLSTM, self-attention, and channel-attention) for EEG-based emotion recognition using time-frequency representations.
- Reduced EEG channels from 32 to 10 while maintaining within 99% of full-channel accuracy; published multiple peer-reviewed papers from this work.

TEACHING EXPERIENCE

- **Graduate Teaching Assistant, University of Maryland (August 2024 - May 2025)**
Courses: ENEE408M: Capstone Design Project: Embedded Software Design (Git, Linux, Bash, C++); ENEE 245: Digital Circuits and Systems Laboratory (Vivado, Verilog HDL, FPGA Programming, Digital Logic Design)
- **Lecturer, Department of EECE, Military Institute of Science and Technology, Dhaka (August 2022- July 2023)**
Courses Taught: Electrical Properties of Materials, Control Systems Laboratory, Communication Systems I Laboratory, Electronic Circuits II Laboratory

PUBLICATIONS

- J1. O. Saha et al.**, “MPNET: A Lightweight Deep Learning Framework for Melt Pool Anomaly Detection in Laser Powder Bed Fusion via Center-Based Representation Learning”, *Journal of Manufacturing Processes* (Under Review)
- J2. O. Saha et al.**, “Automatic Emotion Recognition from Multi-Band EEG Data Based on a Deep Learning Scheme with Effective Channel Attention”, *IEEE Access*, 2022.
- J3. M. S. Mahmud, S. A. Fattah, M. Saquib, O. Saha**, “Emotion Recognition with Reduced Channels Using CWT Based EEG Feature Representation and a CNN Classifier”, *Biomedical Physics and Engineering Express*, 2024.
- C1. O. Saha et al.**, “TCRR: A Clustering-Based Channel Quality Prediction Framework in Industrial Wireless Communication”, *IEEE ICCNS*, Bucharest, Romania, 2026
- C2. O. Saha et al.**, “A Multi-Model Based Ensembling Approach to Detect COVID-19 from Chest X-ray Images”, *IEEE TENCON*, Osaka, Japan, 2020.
- C3. M. S. Mahmud, O. Saha**, “An Efficient Bidirectional LSTM-Based Deep Neural Network for Automatic Emotion Recognition Using EEG Signal”, *ICECE*, 2022.
- C4. J. Tasnim, B. Halder, O. Saha, T. I. Mahmud, S. Ahmed**, “DenseLinkNet: A Deep Convolutional Architecture for Automatic Segmentation of Corneal Endothelial Cell”, *ICECE*, 2022.
- C5. O. Saha et al.**, “DEEPSATTNET: An Efficient Deep Neural Network with Self-Attention Mechanism for Emotion Recognition Utilizing EEG Signal”, *IEEE WIECON-ECE*, India, 2022. (**Best Paper Award**)
- C6. O. Saha et al.**, “Automatic Emotion Recognition from EEG Signal Utilizing Wavelet Packet Node Reconstruction and a CNN Classifier”, *ICCIT*, 2023.

RELEVANT COURSEWORK

Advanced Digital Signal Processing, Estimation and Detection Theory, Natural Language Processing, Multimodal Computer Vision, Information Theory, Fundamentals of Computer Engineering, Compilers and Optimization, Random Signal and Processes, Microprocessors and Embedded Systems